

Lucian Cheng

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Education

McMaster University

Exp. May 2026

Bachelor of Computer Engineering and Biomedical Engineering Co-op

Hamilton, ON

- **GPA: 4.0/4.0**, Provost Honour Roll 2022, 2023, Top 1% of Engineering Cohort
- **Courses:** Data Structures & Algorithms (C++), Software Development, Mathematics, Statistics, Computer Architecture, Operating Systems, Computer Systems Software, Embedded Systems

Professional Experience

Meta

May 2025 - Aug 2025

Software Engineer Intern

Menlo Park, CA

- Created a CLI querying system on the counting and structured data core infra team for **20+** counter types with **C++ / Python**, impacting **1,300+** infra engineers and reducing SEV propagation time by **50%**
- Built a raw cache server state global host retrieval system for **40,000+** servers across **40+** data centers
- Developed production API to implement infrastructure for regional-based querying, handling **trillions** of counter requests daily
- Launched a runtime config failover system using **Hack** that prevents SEV 0s, saving the company **\$90,000,000+** per incident

McMaster EcoCAR

Oct 2022 - Present

CAVs Software Engineer

Hamilton, ON

- Delivered SAE Level 2 vehicle autonomy code in collaboration with **5** cross-functional teams, ensuring compliance with safety standards and meeting deadlines by **20%** faster
- Developed **Python** algorithms for vehicle identification, improving detection accuracy by **15%** on **100+** vehicles

Bluewrist Inc.

May 2024 - Aug 2024

Software Engineer Intern

Markham, ON

- Led a **5-person** team to develop a large-scale full-stack ML application with minimal guidance, designing stakeholder diagrams using **VB**, **Python**, and Agile methodologies
- Implemented 2D anomaly detection models with **98%** accuracy using **PyTorch**, enabling scalable deployment
- Developed an automated image-tiling algorithm leveraging **CUDA** to accelerate ML model training by distributing tasks across GPU cores, increasing system efficiency by **95%**

Bluewrist Inc.

May 2023 - Aug 2023

Software Engineer Intern

Markham, ON

- Implemented the PointNet++ CNN Machine Learning Artificial Intelligence model in PyTorch for part segmentation of computer vision from 3D point cloud data, resulting in a **4-week** time reduction across **10 people**
- Optimized runtime by **86.4% (38.1s)** per **250,000** points using a **C++** inference program with LibTorch & CMake, enabling faster real-time processing and reducing downtime in production systems and internal tools

McMaster University

Sep 2023 - Dec 2023

Physics and Circuits Teaching Assistant

Hamilton, ON

- Mentored **8+** students, improving their grades by **20%** while enhancing critical thinking and problem-solving abilities.

Projects

LeetBuddy | React, TailwindCSS, Express, Redis, Docker, Google Gemini | 

- Created a Google Chrome extension LeetCode AI Assistant, providing intelligent learning help to over **3,000+** active users

Jaylofadez | React, Express, NodeJS, MongoDB, AWS, GCP | 

- Developed and deployed a scalable cloud-native social media platform for **100+** users, leveraging **AWS S3** for scalable asset storage and optimizing backend performance to handle high-traffic conditions, generating **\$600+** in profit

Technical Skills

Programming: Python, C++, JavaScript, TypeScript, Java, Swift, HTML, CSS, C, Verilog, SQL, Hack

Technologies: React, Node, Express, MongoDB, Spring Boot, Flask, .NET, CUDA, PyTorch, Docker, MATLAB

Tools and Platforms: Git, Github, JIRA, SVN, Linux, AWS (EC2, S3), GCP, PostgreSQL

Concepts: Distributed Systems, Automation Tools, Life Cycle, Reliability, Performance Optimization, Rest API, OOP